NEXUM SUUM
Incorporated

Boiler Water Chem Instructions

General Notes:
Follow proper PPE and safety procedures per SDS.
Always use clean, labeled sample containers.
Conduct tests in the following order: Buffer → Indicator → Titrant.
1 mL = 0.06 fl oz.

1. Alkalinity (OH) Test

Purpose: Determines the presence of hydroxide alkalinity in boiler water.
Sample Volume: 25 mL of cooled boiler water sample
Reagents Needed:
20 drops (1 mL) of 20% Barium Chloride
2 drops Phenolphthalein Indicator
Standard HCl Titrant
Procedure:
Pour 25 mL of water sample into a clean test beaker.
Add 20 drops of Barium Chloride and 2 drops of Phenolphthalein.
Swirl. If pink color appears, alkalinity is present.
Titrate slowly with HCl until the pink color disappears.
Record number of drops of HCl used.
Multiply drops by factor provided on reagent kit to get ppm as $CaCO_3$.
Color Change: Colorless → Pink → Colorless

2. Alkalinity (P & M) Test

Purpose: Measures both phenolphthalein (P) and methyl orange (M) alkalinity.
Reagents Needed:
Alkalinity Titrant (Low or High, based on expected range)
3 drops pH Indicator
3 drops Methyl Orange Indicator
Procedure:
Collect sample (volume based on kit instructions).
Add 3 drops of pH Indicator. Titrate with alkalinity titrant until color disappears. Record as P-Alk.
Add 3 drops of Methyl Orange. Continue titration until yellow/orange endpoint is reached. Record as M-Alk.
Interpretation: P-Alk = drops to clear; M-Alk = total drops after orange endpoint.
Color Change: Colorless → Pink → Clear → Yellow → Orange

3. Hardness (Trace Test)

Purpose: Detects trace levels of calcium and magnesium.
Reagents Needed:
5 drops Hardness Buffer
1 scoop Hardness Indicator Powder
Trace Hardness Titrant
Procedure:
Fill test beaker with sample as per test kit.
Add 5 drops buffer and 1 scoop indicator powder.
Swirl gently. Sample will turn red if hardness is present.
Titrate with hardness titrant until the color changes to blue.
Multiply total drops by 5 to get ppm hardness.
Color Change: Colorless → Red → Blue

4. Sulfite Test

Purpose: Measures sulfite residual used for oxygen scavenging.
Important: Sample must be tested immediately after collection and cooled to room temperature.
Reagents Needed:
1 drop pH Indicator
Sulfite Titrant
Procedure:
Add 1 drop of pH Indicator to cooled sample.
Swirl, then begin titrating with Sulfite Titrant until the pink color disappears.
Count the number of drops.
Multiply drops by 5 to calculate ppm sulfite.
Color Change: Colorless → Pink → Colorless